

Chia-Hung Kuo

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Data Scientist | Causal Inference & Experimentation

PhD Candidate in Economics from Michigan State University, specializing in causal inference, experimentation, and A/B testing on datasets exceeding 13M users. Econometrics training means the first question is whether the effect is causal, not whether the model fits. Currently building production statistical systems for an NGO in Taiwan where analysis directly drives annual government funding allocations to universities.

Experience

Taiwan Assessment and Evaluation Association, Taipei, Taiwan (Remote)

Data Science Consultant

09/2024-Current

- Modernized data infrastructure from legacy FTP servers to GCP (GCS, BigQuery) using Python, SQL, and dbt; reduced full analysis cycle from ad hoc to one day
- Built two anomaly detection systems informing government funding decisions across 1,600+ annual evaluations: a time-series regression flagging irregular department scoring patterns, and a latent variable model exploiting reviewer overlap as identification structure to separate genuine quality signals from systematic reviewer bias; delivered via Tableau to non-technical government stakeholders

Michigan State University, East Lansing, Michigan

Lecturer & Teaching Assistant

09/2018-06/2024

- Taught and assisted microeconomics, macroeconomics, international finance courses ranging from 300+ student lectures to 30-person seminars
- Excellence in Teaching Award (2022, awarded to one PhD student annually across the department); Honorable Mention, Peter Schmidt Award (2019; ranked 2nd in PhD cohort qualifying exams)

Projects & Research Papers

Multi-Target Stacking: A Novel Ensemble Framework

Current

- Building ensemble framework that diversifies the prediction target rather than the algorithm: 900+ base models on nested breakout thresholds across 14M+ minute-level records, combined by a tree-based meta-learner
- Designed monthly walk-forward validation pipeline with Optuna hyperparameter tuning and MLflow experiment tracking; built to reduce month-to-month variance in PR-AUC

A/B Testing with Uplift Modeling and Causal Decomposition | [Live App](#)

11/2025

- Implemented S/T/X/R meta-learners using EconML and XGBoost on 13.9M-user RCT; X-Learner best performer (Qini: 0.1753), top decile converting at 5x control rate
- Identified that CATE conflates treatment delivery with treatment response under 96.4% non-compliance; decomposed via IV into reachability $P(E)$ x responsiveness $LATE(X)$, recovering per-user causal estimates that enable differentiated targeting strategy

Research Papers

2021-2024

- Applied DiD-IV quasi-experiment exploiting China's 1986 Compulsory Schooling Law (25 provinces, 16,500+ individuals); estimated 0.84 SD effect concentrated among policy compliers
- Built structural discrete choice model (MLE) on NLSY97 (~9,000 individuals, 20 years) to simulate counterfactual migration decisions; decomposed migration reduction into preference vs. cost channels

Education

PhD Candidate (ABD) in Economics, Michigan State University, East Lansing, MI

06/2024

MSc in Economics, University of Bonn, Bonn, Germany

06/2018

BSc in Quantitative Finance, National Tsing Hua University, Hsinchu, Taiwan

06/2014

Technical Skills

Python, SQL, R, Stata, MATLAB, Git, scikit-learn, XGBoost, EconML, GCP, AWS, dbt, MLflow, Docker